B. Shyam

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51 / 1

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I B Shyam Yadav (192421176) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

B. Shyam Yadav
Signature of the student:

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CO1-AT1

1/ a/ "MEET AT NOON" using ceaser cipher Encrypt & Decrypt.

original letter	shift+4	Encrypted letter
M	N→O→P→Q	Q
E	F→G→H→I	I
E	F→G→H→I	I
T	U→V→W→X	X
A	B→C→D→E	E
T	U→V→W→X	X
N	O→P→Q→R	R
O	P→Q→R→S	S
O	P→Q→R→S	S
N	O→P→Q→R	R

Encrypted message: QIIX EX RSSR

b/ Authorized member. Shifts each encrypted letter four position backward in the alphabet.

MEET AT NOON → QIIX EX RSSR

2,

a, The most likely original message is

THIS IS A SECRET

message appears to use the Atbash cipher which replaces each letter with its opposite letter in the Alphabet.

b, A is replaced by Z, B by Y & C by X

Original Letter: T H I S I S A S E C R E T

Encrypted Letter: G S R H R H Z H X V I G

3,

a, Vigenere cipher uses a repeating keyword to encrypt each

Plaintext: HELLO

Keyword: KEYKE

$$c = (p + k) \bmod 26$$

H	K	(7+10) mod 26 = 17	R
E	E	(4+4) mod 26 = 8	I
L	Y	(11+24) mod 26 = 9	J
L	K	(11+10) mod 26 = 21	V
O	E	(14+4) mod 26 = 18	S

b/

- Key is repeated to match the message length.
- Plaintext letter is combined with its corresponding keyword letter.
- Different keyword letters produce different shifts in Alphabet.
- Plaintext letter taken paired with different key letter.
- Receiver uses the same keyword to decrypt the ciphertext.

4/

(b) Encrypt the secret message before embedding it. AES use to remain the unreadable without the decryption key.

Use keyed steganograph: data using secret key and embedding patten to make difficult for unauthorized users.

(a) (i) metadata Analysis: Image property file detail & metadata for suspicious information.

(ii) File Size Analysis: image file size is usually large for its Dimension & Quality.

(iii) Pixel Analysis: Pixel value & color channels.

(iv) LSB Analysis: least significant bits of Pixel.

(v) Statistical Analysis: compare Pixel value distribution to identify possible.

(vi) Steganography: To identify and extract hidden data when the embedding method is known.

5,

(a)

Ciphertext :

WE RCL TEERD SOEEFE AOCAIVDEN.

WE ARE DISCOVERES FLEE AT ONCE

This phrase is a well known Example of transposition.

(b)

* Preserve letter frequency.
* Limited security
* Susceptible to attacks -
* No modern security guarantees.
* Key recovery is easier.

(c)

Replace AES 256 Authorized Authentication Encryption such as AES-GCM.

Benefits :

* Provide strong confidentiality.
* Detects unauthorized modification of encrypted message.
* Cryptographic keys rather than relying on simple letter arrangements.
* Suitable for protecting sensitive corporates data. when implemented correctly with secures key management & unique nonce.